MINI PROJECT 2

🚦 Urban Traffic Congestion Analysis & Risk Prediction System

◆ Problem Statement

Urban cities face increasing traffic congestion and accident risks. Traffic departments need a system to **analyze congestion patterns, peak hours, accident-prone zones, and predict future congestion levels**.

🎯 Objectives

- Analyze traffic data by time, location, and weather
- Identify peak congestion patterns
- Predict congestion level
- Visualize accident risk zones
- Build an interactive Streamlit dashboard

📋 Tasks to be Performed

1. Load traffic and accident data
2. Data preprocessing
3. Peak hour analysis
4. Congestion pattern identification
5. Risk prediction using ML
6. Geospatial and time-based visualization

📈 Visualizations & Patterns

- Line Chart → Traffic volume vs time
- Bar Chart → Peak congestion hours

- Heatmap → Accident-prone areas
- Area Chart → Congestion intensity
- Risk Score Indicator

🛠 Technologies Used

- Python
- Pandas, NumPy
- Matplotlib
- Scikit-learn
- Streamlit

⍰ Solution Approach

ML Models

- Logistic Regression / Decision Tree for congestion risk

Pattern Detection

- Morning & evening peak hours
- Weekday vs weekend traffic differences
- High-risk accident zones